METRIC

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MILITARY STANDARD

INORGANIC BASES AND BASIC ANHYDRIDES, TECHNICAL GRADE (METRIC)



AMSC N/A

FSC 6810

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FOREWORD

Inorganic Bases and Basic Anhydrides, Technical Grade

1. This Military Standard is approved to use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to : Commander, U.S. Army Edgewood Research, Development and Engineering center, Attn: SMCCR-PET-S, Aberdeen Proving Ground, MD 21010-5423, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

3. This standard is approved for use by all Departments and Agencies of the Department of Defense in the selection of items for application. It is intended to prevent the entry of unnecessary items (sizes, types, varieties) into the Department of Defense logistics system. This document is not intended to restrict any service in selecting new items resulting from state-of-the-art changes.

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1. SCOPE

1.1 Coverage. This standard is a presentation of nomenclature, Chemical Abstracts Service Registry Numbers, formulas, physical and chemical properties, specification requirements, military and typical commercial uses, safety information, storage information and disposal information for inorganic bases and basic anhydrides, technical grade. This standard does not include all of the items represented by the title or all those items which are commercially available. It does contain items preferred for use in the selection of inorganic bases and basic anhydrides, technical grade, for application by the Department of Defense.

1.2 Application. Inorganic bases and basic anhydrides, technical grade, are used mainly as cleaning agents, in neutralizing acid wastes, water treatment, as dehydrating agents, in electroplating, gas absorption and as anti-fouling agents and in pyrotechnic compositions. Inorganic bases and basic anhydrides are also used as chemical compounds and in the formulation of chemical products for a variety of applications.

1.3 Classification. The items in this standard are classified on the basis of chemical composition as inorganic bases and basic anhydrides, technical grade.

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2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

O-A 451	- Ammonium Hydroxide, Technical
O-C-110	- Calcium Hydroxide, Technical
O-P-566	- Potassium Hydroxide, Technical
O-S-598	- Sodium Hydroxide, Technical
PPP-C-2020	- Chemicals, Liquid, Dry and Paste, Packaging of

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MIL-P-11751	- Potassium Hydroxide, Solution
MIL-C-12056	- Calcium Oxide, Technical (Metric)
MIL-C-13600	- Cupric Oxide
MIL-L-20213	- Lithium Hydroxide (LiOH), Technical
MIL-S-50005	- Stannic Oxide, Technical

STANDARDS

FEDERAL

FED-STD-313	- Material Safety Data Sheets, Preparation And The Submission Of

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MIL-STD-12	- Abbreviations For Use On Drawings, And In Specifications, Standards And Technical Documents

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available for the DODSSP-Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

Title 29	Department of Labor, Occupational Safety and Health Agency (OSHA); General Industry Standards And Interpretations
Title 40	Environmental Protection Agency (EPA); Hazardous Waste And Consolidated Permit Regulations
Title 49	Department Of Transportation; Hazardous Materials Regulations

DOD 4145.19-R-1	Storage And Materials Handling
DOD 4160.21-M	Defense Utilization And Disposal Manual
DOD 6050.5-LR	DOD Hazardous Materials Information System, Hazardous Item Listing.
TB MED 502 (DLAM 1000.2)	Occupational And Environmental Health Respiratory Protection Program.
TB MED 506	Occupational And Environmental Health Occupational Vision
TM 38-250	Packaging, Materials Handling-Preparation Of Hazardous Materials For Military Air Shipment.

NATIONAL INSTITUTE FOR OCCUPATIONAL SAFETY AND HEALTH (NIOSH)

Registry of Toxic Effects of Chemical Substances
Recommendation for Environmental Exposure Limits

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the

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documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

AMERICAN CONFERENCE OF GOVERNMENTAL INDUSTRIAL HYGIENISTS (ACGIH)

Threshold Limit Values (TLVs) for Chemical Substances in the Work Environment Adopted by American Conference of Governmental Industrial Hygienists (ACGIH) with Intended Changes.

(Application for copies should be addressed to American Conference of Governmental Industrial Hygienists, 6500 Glenway Avenue, Bldg D-7, Cincinnati, OH 45211-4438.)

ASTM

ASTM D 456	Standard Specification for Caustic Soda (Anhydrous)
ASTM E 11	Standard Specification for Wire-Cloth Sieves For Testing Purposes
ASTM E 291	Standard Methods for Chemical Analysis Of Caustic Soda And Caustic Potash (Sodium Hydroxide and Potassium Hydroxide)
ASTM E 323	Standard Specification For Perforated - Plate Sieves For Testing Purposes
ASTM E 380	Standard For Metric Practice

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

CHEMICAL ABSTRACTS SERVICE REGISTRY

The Chemical Abstracts Service (CAS) Registry number provides a key to access the wealth of information available through CAS.

(For details on how to search the CAS files and database using the registry number, contact Chemical Abstracts Services, Customer Service, P.O. Box 3012, Columbus, OH 43210.)

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

National Fire Codes

(Application for copies should be addressed to National Fire Protection Association, Battery March Park, Quincy, MA 02269.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. DEFINITIONS

3.1 Definitions of Technical Terms.

3.1.1 Baume.

For liquids lighter than water:

$$°\text{Baume} = \left[\frac{140}{\text{Specific gravity } 60°F/60°F}\right] - 130$$

For liquids heavier than water:

$$°\text{Baume} = 145 \left[\frac{140}{\text{Specific gravity } 60°F/60°F}\right]$$

Specific gravity 60°F/60°F = Ratio of density of liquid at 60°F to density of distilled water at 60°F.

3.1.2 Oxidizing agent - A substance that gains electrons as a result of an oxidation-reduction reaction. It causes an increase in the oxidation state of another substance.

3.1.3 Reducing agent - A substance that loses electrons during an oxidation-reduction reaction. It causes a decrease in the oxidation state of another substance.

3.2 Definitions of abbreviations. The use of abbreviations shall be in accordance with MIL-STD-12 where applicable. Metric system abbreviations and symbols shall be in accordance with ASTM E 380.

Additional abbreviation are as follows:

FW - Formula Weight
CAS - Chemical Abstracts Service Registry Number
TLV - Threshold Limit Value
PEL - Permissible Exposure Limit
LEL - Lower Explosive Limit
UEL - Upper Explosive Limit

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4. GENERAL REQUIREMENTS

4.1 Packaging data and labeling. All chemicals included in this standard shall be packaged in accordance with Federal Specification PPP-C-2020 and all applicable documents referenced therein. Shipping containers shall be labeled in accordance with current Department of Transportation (DOT) Hazardous Materials Regulations applicable to each chemical; when shipping by military aircraft, the requirements of TM 38-250 shall apply. In addition, each item shall be packaged and labeled as specified in the applicable contract or order. All labels shall also comply with Hazard Communication Standard, 29 CFR 1910.1200 (f).

4.2 Hazardous materials information. DOD 6050.5, DOD Hazardous Materials Information System (HMIS) acquires, reviews, stores, and disseminates Material Safety Data Sheet (MSDS) information for all hazardous materials used by DOD. The contractual acquisition of an MSDS is accomplished through use of Federal Acquisition Regulation, paragraph 52.223-3, Hazardous Material Identification and Material Safety Data. The MSDS is prepared in accordance with the instructions in FED-STD-313 and shall comply with the requirements of Hazard Communication Standard, 29 CFR 1910.1200 (g).

4.3 Safety.

4.3.1 Personal protective measures. The necessary respiratory, eye and skin protection to be used when handling chemicals shall be prescribed by the responsible installation industrial hygiene, medical and safety authorities.

4.3.1.1 Respiratory protection. Respirators, approved by the National Institute for Occupational Safety and Health (NIOSH) or the Mine Safety and Health Administration (MSHA) or by particular respiratory schedules of the Bureau of Mines (BM) for the compounds being used, may be employed for intermittent, non-routine exposure (not exceeding one hour per week), when the installation medical authority determines that there are no feasible engineering or work practice controls, during interim periods when engineering controls are being designed and/or installed, during emergencies, or for supplementing other control measures (refer to TB MED 502 or DLAM 1000.2). Ventilation containment, process controls, or other feasible engineering controls shall be adequate to remove hazardous concentrations. Respiratory protection shall not be used in place of feasible engineering controls.

4.3.1.2 Skin protection. Personnel using these compounds shall be provided with and required to use impervious gloves, sleeves, aprons, and boots whenever indicated. Protective creams and ointments commonly known as "barrier creams" may be of value in certain cases. However, barrier creams shall not be used to replace

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protective clothing. In case of contact with the skin, wash affected areas thoroughly with water. Eye lavages and emergency showers shall be located where there is a potential for direct contact with harmful chemicals.

4.3.1.3 Face and eye protection. Personnel using these compounds shall be provided with and required to wear chemical splash-proof safety goggles. In addition, face shields shall be provided and worn over the goggles if splashing could occur. In case of contact with the eyes, immediately irrigate with copious amounts of water for at least 20-30 minutes, and obtain medical attention. (Refer to TB MED 506.)

4.3.1.4 Training. Employers shall provide employees with training and information, including MSDSs, on all chemical items in their work area, in accordance with 29 CFR 1910.1200 (h), to ensure that employees know potential hazards of the chemicals with which they come in contact and the symptoms of exposure, as well as how these chemicals affect the body and bodily functions. Employees shall be adequately trained to render first aid.

4.3.1.5 Exercises. Participation in training exercises shall be stressed to demonstrate skills in the use of personal protective equipment and emergency response equipment.

4.3.2 Storage conditions. DOD 4145.19-R-1 describes general storage practices and requirements for hazardous materials in the DOD supply system. Specific requirements provided in the following paragraphs are supplementary in nature and shall be observed in consonance with the DOD storage regulations.

4.3.2.1 Flammable, combustible, pyrophoric and ignitable materials. A flammable material is generally any solid, liquid, vapor or gas that ignites easily and burns rapidly. Combustible materials are generally those that are difficult to ignite and burn slowly. The DOT, in Part 173, Subpart D, Section 173.115 of 49 CFR, defines a flammable liquid as one having a closed cup flash point below 37.8°C (100°F). A combustible liquid is defined, by DOT in the above reference, as one having a closed cup flash point at or above 37.8°C (100°F) and below 93.3°C (200°F). A pyrophoric liquid is defined, by DOT in the above reference, as one that ignites spontaneously in dry or moist air at or below 54.5°C (130°F). Materials with flash points of 93.3°C (200°F) or higher are to be considered as burnable. The Environmental Protection Agency (EPA), in Part 261, Subpart C, Section 261.21 of 40 CFR*, designates the criteria for flammable and combustible materials and oxidizers that exhibit the characteristic of

*Refers only to materials that have become waste.

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ignitability (I). Liquids with closed cup flash points of less than 60°C (140°F) are defined by EPA as ignitable. The autoignition point (temperature) of a substance is generally defined as the minimum temperature required to initiate or cause self-sustained combustion in the absence of a spark or flame. Materials that ignite easily under normal industrial conditions are considered to be dangerous fire hazards. Such materials shall be stored in a manner to prevent ignition and combustion. Easily ignitable substances, such as reducing agents, shall be kept away from strong oxidizing agents. All containers shall be tightly sealed. It is important to provide adequate ventilation in storage areas, and to locate the storage areas of these items away form fire hazards. Ample fire-control equipment shall be easily accessible. Storage buildings, rooms and cabinets shall comply with provisions of the National Fire Codes. The building shall be electrically grounded and signs posted to prevent the lighting of matches or smoking in the area. Flammable storage areas shall be equipped with smoke or fire detection equipment.

4.3.2.2 Water-sensitive fire and explosive hazardous materials. These are materials that react on contact with water or steam to ignite or evolve heat or explosive gases. Such materials exhibit the characteristic of reactivity (R) as designated by the EPA in Section 261.23 of the above reference.* These materials shall be stored in well-ventilated, cool, dry areas. All containers shall be tightly sealed. These materials are a fire hazard in contact with water or moisture; therefore, it is essential that no sprinkler be used. Otherwise, the building shall conform to that required for storage of flammable materials. The building shall be waterproof, located on high ground, separated from other storage areas and meet National Fire Codes.

4.3.2.3 Incompatible materials. Materials that are chemically incompatible shall be segregated in the storage of both serviceable and unserviceable items. The degree of segregation will depend upon DOD 4145.19-R-1 and local supplementary requirements that insure safe storage conditions. Hazardous storage compatibility codes are provided in the HMIS referred to in 4.2.

4.3.3 Chemical hazardous exposure limits. Chemical hazardous exposure limits for airborne concentrations of substances are obtained from the current TLVs Threshold Limit Values for Chemical Substances in the Work Environment adopted by the American Conference of Governmental Industrial Hygienists (ACGIH); current Occupational Safety and Health Administration (OSHA) Permissible Exposure Limits (PEL), 29 CFR, Part 1910, Section 1910.1000; and NIOSH Recommended Exposure Limits. Such information is also shown in MSDSs and the HMIS referred to in 4.2.

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Carcinogenic substances are listed by OSHA in Category I for confirmed carcinogens, and in Category II for suspected carcinogens. Category II substances have standard exposure limits set to prevent acute or chronic effects.

4.3.4 Toxicity. Toxicity information for chemical compounds is available from various publications and from MSDSs, which are collected in DOD 6050.5-LR Hazardous Materials Information System.

4.3.4.1 EPA Toxic (T). Some chemical compounds have been designated by the EPA as toxic (T) in accordance with the criteria shown in Part 261, Subpart B, Section 261.11(a)(3) of 40 CFR.* Some commercial chemical products are listed as toxic under Subpart D, Section 261.33 (f).

4.3.4.2 EPA Acute Hazardous Toxicity (H). Some chemical compounds have been designated by the EPA as acutely hazardous (H) in toxicity in accordance with the criteria shown in Subpart B, Section 261.11(a)(2) of 40 CFR.* Some commercial chemical products are listed as acutely hazardous in toxicity under Subpart D, Section 261.33(e).

4.3.4.3 EPA Toxicity Characteristic Leaching Procedure (TCLP). This is a procedure used to determine the properties of a waste that are directly related to the potential of the waste to pose a hazard to groundwater when disposed of in a landfill environment. It replaces the EPA EP toxicity test. The TCLP procedure is contained in EPA Method No. 1311, SW-846. Some 40 chemical contaminants have been designated by the EPA as being a hazard to groundwater in a landfill environment. (See Section 261.24 of 40 CFR.)*

4.4 Pollution and disposal.

4.4.1 Pollution potential. All items described in this standard shall be assumed to have a pollution potential. However, to minimize this potential, the proper use, storage and disposal methods shall be strictly followed.

4.4.2 Disposal of excess or unserviceable material. To minimize disposal problems, it is recommended that no more than a one year's supply of each item listed in this standard be stocked. When stocks have been declared excess or unserviceable, they will be disposed of in accordance with the Defense Utilization and Disposal Manual, DOD 4160.21-M, and applicable DOD Policy Memoranda. Guidance

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can be obtained from your servicing Defense Reutilization and Marketing Office (DRMO) on procedures required for proper reporting and turn-in.

4.4.3 Disposal and storage of hazardous wastes. Items are classified and managed as hazardous wastes as defined by the Resource Conservation and Recovery Act (RCRA) (Public Law 94-580). Items have been identified as meeting the characteristics (i.e., ignitable, corrosive, reactive) or are subject to Toxicity Characteristic Leaching Procedure (TCLP) or are listed (i.e., toxic or acutely hazardous) according to Identification and Listing of Hazardous Waste, Part 261, 40 CFR; or have been determined to be hazardous wastes by declaration of the Defense Reutilization and Marketing Service (DRMS) in accordance with procedures set forth in DOD 4160.21-M. Disposal of such items shall be managed in accordance with the Installation Environmental Office, the DRMO, or the Safety and Health Office to insure proper reporting of disposal and treatment actions to the US EPA and State, and shall be managed in accordance with Federal, State and local laws. The three main disposal methods are turn-in to the DRMO, on-post disposal by installation personnel, or disposal by commercial contract. Hazardous wastes that cannot be disposed of as stated in 4.4.3.2 shall be stored under environmentally safe conditions until suitable methods of disposal are determined. Short-term storage (less than 90 days) requires proper containment (i.e., packaging and facilities) in accordance with 40 CFR Section 262.34. Long-term storage (greater than 90 days) requires permitting by the EPA or by the state under Public Law 94-580 (RCRA), in compliance with the requirements of 40 CFR, Parts 264 and 265. Physical custody will be accomplished by the activity with conforming storage or most nearly conforming storage. When physical custody is in question, the Post Commander will make the final decision. In all cases where the wastes are to be collected, stored, transported and disposed of at a state or local permitted disposal facility, the identity and description of the waste shall be maintained and recorded in accordance with Part 262 of the above reference. Transportation of the waste must be in accordance with Part 263 of the above reference, Standards Applicable to Transporters of Hazardous Waste. State and/or local regulations may be more stringent than Federal requirements. It is essential that the installation Environmental Office review all disposal actions for compliance with all applicable regulations.

4.4.3.1 Cleanup of liquid spills. To control the migration of spilled or leaking liquids, dike around the item with an inert, dry absorbent (e.g., clay or vermiculite) or follow installation spill plans (Spill Prevention Control and Countermeasure Plan and Installations Spill Contingency Plan). Control entry to the spill site and segregate salvageable materials away from the spill area. Initiate waste cleanup operations immediately in accordance with local procedures. The residue shall be safely handled and transported to an approved or permitted disposal or storage facility. Packaging, labeling, transportation and record-keeping requirements for this waste material are

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determined by the appropriate Federal and State agencies and local procedures. It is recommended that all activities involving disposal preparation and transportation to commercial facilities be properly coordinated with the appropriate Federal and State agencies responsible for health and environmental aspects of hazardous materials. It is imperative that the proper description of waste accompany the packaged item at all times. Final disposal of the waste item shall be accomplished by reutilization, transfer, donation or sales by DRMS in accordance with DOD 4160.21-M or by ultimate disposal as described in 4.4.3.2. Spill residue, including contaminants, to be turned into the DRMO shall first be properly identified, containerized, and labeled. For large scale spills that grossly contaminate the environment, the Chemical Transportation Emergency Center (CHEMTREC) can be called for assistance (1-800-424-9300). Applicable procedures of the local spill control plan shall be followed. Necessary respiratory, eye, and skin protection measures are to be used while performing cleanup operations.

4.4.3.2 Ultimate disposal. Ultimate disposal shall be accomplished at a permitted or approved hazardous waste treatment or disposal facility designated by the Installation Environmental Office, DRMO, or Safety and Health Offices.

4.4.4 DISCLAIMER. RECOMMENDED DISPOSAL INSTRUCTIONS ARE FORMULATED FOR USE BY ELEMENTS OF THE DEPARTMENT OF DEFENSE. THE UNITED STATES OF AMERICA IN NO MANNER WHATSOEVER, EITHER EXPLICITLY OR IMPLICITLY, WARRANTS, STATES, OR INTENDS SAID INSTRUCTIONS TO HAVE ANY APPLICATION, USE OR VIABILITY BY OR TO ANY PERSON OR PERSONS CONTRACTING OUTSIDE THE DEPARTMENT OF DEFENSE OR ANY PERSON OR PERSONS CONTRACTING WITH ANY INSTRUMENTALITY OF THE UNITED STATES OF AMERICA, AND DISCLAIMS ALL LIABILITY FOR SUCH USE. ANY PERSON USING THESE INSTRUCTIONS WHO IS NOT A MILITARY OR CIVILIAN EMPLOYEE OF THE UNITED STATES OF AMERICA SHOULD SEEK COMPETENT PROFESSIONAL ADVICE TO VERIFY AND ASSUME RESPONSIBILITY FOR THE SUITABILITY OF THESE INSTRUCTIONS TO HIS PARTICULAR SITUATION, REGARDLESS OF SIMILARITY TO A CORRESPONDING DEPARTMENT OF DEFENSE OR OTHER GOVERNMENT SITUATION.

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5. DETAILED REQUIREMENTS

5.1 Name. AMMONIUM HYDROXIDE, TECHNICAL NH_4OH FW 35.05
Aqua ammonia
Ammoniacal liquor CAS 1336-21-6
Ammonium hydrate
Ammonia water

5.1.1 Technical description. Ammonium hydroxide (pure) exists only in solution at ordinary temperatures. It is a colorless liquid having a strong pungent odor. Its melting point is -77°C. It is soluble in cold water.

5.1.2 Specification. Federal, O-A-451, Ammonium Hydroxide, Technical.

This specification covers four types of ammonium hydroxide, technical:

Type I - 26° Baume (27 to 30 percent as NH_3)
Type II - 21° Baume (19 to 21 percent as NH_3)
Type III - 16° Baume (9 to 10 percent as NH_3)
Type IV - 14° Baume (6 to 7 percent as NH_3)

5.1.3 Requirements. The Federal Specification requirements for ammonium hydroxide, technical, are shown in Table I.

TABLE I. Ammonium hydroxide, technical requirements.

Requirements	Type I	Type II	Type III	Type IV
Assay (as NH_3), % by wt, max	27-30	19-21	9-10	6-7
Specific gravity at 60°/60°F, max	0.904	0.929	0.936	0.974
Maximum Limit of Impurities, % by wt				
Residue after ignition	0.03	0.03	0.03	0.03
Nonvolatile matter	0.05	0.05	0.05	0.05

Ammonium hydroxide (26° Baume) is commercially available with a maximum density of 0.8974 for 29.4% NH_3 at 15.5°C(60°F).

5.1.4 Use. Ammonium hydroxide, technical, is intended for military use as a cleaning agent, an accelerator in vulcanization, a dry diazo print developer for white print machines and as a laboratory reagent, when this grade is suitable.

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Commercially, it is used in textiles, fertilizers, plastics, pharmaceuticals and lubricants. It is also used in fireproofing wood, ink manufacture, explosives, ceramics, ammonium compounds, saponifying fats and oils, glues, tanning leather and in metal industries.

5.1.5 Safety. Aqueous ammonia exerts a local irritant action; strong solutions cause tissue destruction on contact, whether acting on eyes, skin, mucous membranes, gastrointestinal mucosa, or pulmonary tissue. Concentrated solutions of ammonia remaining in contact with the eyes for even a short time may cause serious ocular damage, which may result in prolonged, severe visual disturbances or permanent scarring of the cornea. The consequences of skin contact with ammonia water vary from a relatively mild dermatitis to severe burns, depending upon the strength of the solution, length of contact, and individual skin sensitivity.

Ammonia gas causes irritation of the eyes, respiratory tract and throat at concentrations as low as 0.05 percent to 0.1 percent. A concentration of 0.2 percent produces convulsive coughing and may be fatal after a short exposure, i.e., less than half an hour. The maximum concentration tolerated by the skin for more than a few seconds is 2.0 percent. Ammonium hydroxide shall be used with adequate ventilation. Inhalation of vapors or mist shall be avoided. Contact with the eyes, skin and clothing shall be avoided. (Refer to 4.3.1)

When concentrated solutions of ammonia are mixed rapidly with sulfuric acid or other strong mineral acids, the mixture becomes boiling hot instantly and may result in an explosion. Mixing with hypochlorites (chlorine bleach), other halogens and sodium hydroxide shall also be avoided. Avoid contact with metals. Heating shall also be avoided. Ammonia decomposes to hydrogen and nitrogen gases above 450°C (842°F) and has a flash point of 60°C (closed cup). Ammonia has a Threshold Limit Value (TLV) of 25 ppm (18 mg/m^3). The Permissible Exposure Limit (PEL) is 50 ppm (35 mg/m^3). Evolved ammonia has a lower explosive limit (LEL) of 16% and an upper explosive limit (UEL) of 25% in air.

5.1.6 Storage. Ammonium hydroxide shall be stored in a cool, well-ventilated place in tightly closed strong glass or plastic containers. (Refer to 4.3.2.)

5.1.7 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices. (Refer to 4.4.)

Ammonium hydroxide is on the EPA's list of hazardous substances and is considered a corrosive substance.

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repeated breathing of dust. Conditions of exposure may necessitate the use of appro

5.2 Name. BARIUM HYDROXIDE, TECHNICAL $Ba(OH)_2.8H_20$ FW 315.48
Barium hydrate
Caustic baryta
CAS 12230-71-6

5.2.1 Technical Description. Barium hydroxide crystallizes with eight molecules of water and exists as transparent crystals or white masses. It rapidly absorbs carbon dioxide from the air to become incompletely soluble in water. It has a density of 2.13, and a melting point of 78°C. One gram dissolves in 11.6 milliliters of water at room temperature or 30 milliliters of boiling water. When heated to 100°C, it loses water and becomes $Ba(OH)_2.H_20$. It is strangely alkaline in aqueous solutions and is slightly soluble in alcohol.

5.2.2 Specification. Manufacturer's requirements. (No Government specification.)

5.2.3 Requirements. Barium hydroxide, technical grade, is commercially available with a typical analysis range as shown in Table II.

TABLE II. Barium hydroxide, technical - typical analysis range.

Requirements	% by wt
	Typical Analysis Range
Barium hydroxide [$Ba(OH_2)$]	95-99
Limits of Impurities	
Barium carbonate ($BaCO_3$)	2.0-50
Insoluble, dilute HCl	0.01
Iron (Fe)	0.1
Strontium (Sr)	0.1-1.2
Calcium (Ca)	0.1-0.2
Silica (Si)	trace
Chloride (Cl)	0.01
Not H_2SO_4 pptd	2.0-5.0
Sulfide (S)	trace
Heavy Metals	5 ppm

5.2.4 Use. Barium hydroxide, technical, is used as an oil and grease additive, in barium soaps, as an alkalizing agent in water, a sulfate removal agent in treatment

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of water and brine, for boiler scale removal and as a purifying agent for caustic soda. It is also used in the sugar industry and in refining animal and vegetable oils.

5.2.5 Safety. Barium hydroxide is toxic by inhalation. Barium hydroxide aqueous solutions are strongly alkaline and therefore irritating to the eyes, skin, and mucous membranes. Prolonged contact with solutions of barium hydroxide has a destructive effect on tissue. Contact with the eyes, either in solid or solution form, can cause damage to the delicate tissue. Inhalation of the dust or mist can cause damage to the upper respiratory tract and to lung tissue, depending on length of exposure. Inhalation of dust or mist shall be avoided. Contact with the eyes, skin and clothing shall be avoided. Barium hydroxide shall be used with adequate ventilation. The TLV for barium, soluble compounds, as Ba is 0.5 mg/m^3. The PEL is the same. (Refer to 4.3.1.)

5.2.6 Storage. Barium hydroxide shall be stored in a cool, dry place in tightly closed containers. (Refer to 4.3.2.)

5.2.7 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, cr Safety and Health Offices. (Refer to 4.4.)

Barium is listed in 40 CFR Section 261.24, Table 1 for toxicity characteristics. Barium hydroxide will exhibit the characteristic of toxicity for barium. Specific State and local regulations may treat barium hydroxide as a corrosive waste. Designations of toxic and corrosive hazardous waste apply only to material declared as waste.

5.3 Name. BARIUM OXIDE, TECHNICAL BaO FW 153.34
Barium monoxide
Barium protoxide CAS 1304-28-5
Calcined baryta

5.3.1 Technical Description. Barium oxide (pure) exists as colorless cubic crystals or a white to yellowish powder with a refractive index of 1.98 and density of 5.72 g/cm^3. Its melting point is 1923°C and boiling point is about 2000°C. Its solubility in water is 3.48 g/100 cm^3 at 20°C and 90.8 g/cm^3 at 100°C, forming barium hydroxide. It is soluble in alcohol, forming barium alcoholate. It is soluble in dilute acids. It is insoluble in acetone.

5.3.2 Specification. Manufacturer's requirements. (No Government specification.)

5.3.3 Requirements. Barium oxide, technical grade, is commercially available with a typical analysis as shown in Table III.

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TABLE III. Barium oxide, technical - typical analysis.

Requirement	% by wt
Assay (BaO), min-max	88-92
Maximum Limits of Impurities	
Barium carbonate ($BaCO_3$), min-max	7-11
Insoluble in HCl	0.45
Iron (as Fe)	0.017
Sodium (as Na)	0.032
Sulfur (as S)	0.03
Silicon dioxide (SiO_2)	0.670
Aluminum (as Al)	0.05

5.3.4 Use. Barium oxide, technical, is used as a dehydrating agent.

5.3.5 Safety. Barium oxide is toxic if ingested or inhaled. It is irritating to the eyes, skin and mucous membranes. Inhalation of dust shall be avoided. Contact with the eyes, skin and clothing shall be avoided. Barium oxide shall be used with adequate ventilation. The TLV for soluble barium compounds (as Ba) is 0.5 mg/m^3. The PEL is the same. (Refer to 4.3.1.)

On contact with water, barium oxides evolve much heat.

5.3.6 Storage. Barium oxide shall be stored in a cool, dry place in tightly closed containers. (Refer to 4.3.2.)

5.3.7 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices. (Refer to 4.4.)

Barium is listed in 40 CFR Section 261.24, Table 1 for toxicity characteristics. Barium oxide will exhibit the characteristic of toxicity for barium. Specific State and local regulations may treat barium oxide as a corrosive waste. Designations of toxic and corrosive hazardous waste apply only to material declared as waste.

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5.4 Name. CALCIUM HYDROXIDE, TECHNICAL $Ca(OH)_2$ FW 74.09
Hydrated lime
Slaked lime CAS 1305-62-0
Calcium hydrate

5.4.1 Technical Description. Calcium hydroxide (pure) exists as colorless hexagonal crystals with a density of 2.24 g/cm^3. It loses water at 580°C and decomposes on boiling. Its solubility in water is 0.185 g/100 cm^3 at 0°C and 0.077 g/100 cm^3 at 100°C. It is soluble in NH_4 salts and insoluble in alcohol. Calcium hydroxide is unstable if subjected to CO_2 in moist air and acids.

5.4.2 Specification. Federal, O-C-110, Calcium Hydroxide, Technical.

5.4.3 Requirements. The Federal Specification chemical and physical requirements for calcium hydroxide, technical, are shown in Table IV and Table V.

TABLE IV. Calcium hydroxide, technical - chemical requirement.

Requirement	% by wt
Assay [$Ca(OH)_2$], min	90.0

TABLE V. Calcium hydroxide, technical - granulation requirements.

U S Standard Sieve 1/	% by wt
Thru 600 micrometer (No. 30), min	99.5
Thru 75 micrometer (No. 200), min	98.0

1/ Standard sieve designation in accordance with ASTM E 11.

5.4.3.1 Appearance. Calcium hydroxide shall be white, dry, finely powdered, and free from lumps or any foreign material that might interfere with the operation of dry-feed equipment.

5.4.4 Use. Calcium hydroxide, technical, is intended for use in the treatment of sewage, industrial water, and potable water, as well as for general use. It is not intended for structural use such as in the manufacture of mortar or plaster.

5.4.5 Safety. Calcium hydroxide is irritating to the eyes, skin and mucous membranes. Eyes and open cuts are particularly vulnerable. Inhalation of dust shall be avoided. Calcium hydroxide shall be used with adequate ventilation. Dusting shall be avoided. The TLV for calcium hydroxide is 5 mg/m^3. There is no PEL referenced for calcium hydroxide. (Refer to 4.3.1.)

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5.4.6 Storage. Calcium hydroxide shall be stored in a cool, dry, well-ventilated place in tightly closed containers away from acids. (Refer to 4.3.2.)

5.4.7 Disposal. For appropriate procedures, contact the Installation Environmental Office, The DRMO, or Safety and Health Offices. (Refer to 4.4.)

Calcium hydroxide is in the EPA inventory under TSCA and meets the criteria for a solid corrosive substance.

5.5 Name. CALCIUM OXIDE, TECHNICAL CaO FW 56.08
Unslaked lime
Burnt lime CAS 1305-78-8
Quicklime

5.5.1 Technical description. Calcium Oxide (pure) exists as colorless, cubic crystals with an index of refraction of 1.838 and a density of 3.25 - 3.38 g/cm^3. Its melting point is 2580°C and boiling point is 2850°C. Its solubility in water is 0.131 g/100 cm^3 at 10°C and 0.07 g/100 cm^3 at 80° with decomposition.

5.5.2 Specification. Military, MIL-C-12056, Calcium Oxide, Technical (Metric).

This specification covers two grades of calcium oxide, technical:

Grade A - 90 percent purity
Grade B - 95 percent purity

5.5.3 Requirements. The Military Specification requirements for calcium oxide, technical, are shown in Table VI and Table VII.

TABLE VI. Calcium oxide, technical - chemical requirements.

Requirements	% by wt			
	Grade A		Grade B	
	Min	Max	Min	Max
Nonvolatile matter	94	--	98	--
Calcium oxide 1/	90	--	95	--
Magnesium oxide 1/	--	--	--	1.0
Iron and aluminum oxides 1/	--	--	--	1.0
Silicon oxide 1/	--	--	--	1.5

1/ On a nonvolatile basis

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TABLE VII. Calcium oxide, technical - granulation requirements.

Particle Size 1/	% by wt			
	Grade A		Grade B	
	Min	Max	Min	Max
Retained on 381. millimeter (1 1/2 inch) square opening	-	0.0	-	-
Retained on 25.4 millimeter (1 inch) square opening	-	8.0	-	-
Retained on 3.35 millimeter (1/8 inch) round opening	95.0	-	-	-
Retained on 5.6 millimeter (0.223 inch)(No. 3 1/2 standard sieve)	-	-	-	0.0
Retained on 1.70 millimeter (0.0661 inch)(No. 12 standard sieve)	-	-	95.0	-

1/ Standard sieve designation in accordance with ASTM E 11 and ASTM E 323 as applicable.

5.5.4 Use. Calcium oxide, technical, is intended for use in steel furnace fluxing, in neutralizing acidic wastes, and in manufacturing bleaching materials.

5.5.5 Safety. Calcium oxide is irritating to the eyes, skin and mucous membranes. Inhalation of dust shall be avoided. Contact with the eyes, skin and clothing shall be avoided. Calcium oxide shall be used with adequate ventilation. The TLV for calcium oxide is 2.0 mg/m^3. The PEL is 5.0 mg/m^3. (Refer to 4.3.1.)

Calcium oxide is not combustible, but contact with water or moisture may generate sufficient heat to ignite combustible materials. It swells when moist and may burst containers. It is not compatible with acids.

5.5.6 Storage. Calcium oxide shall be stored in a dry place in tightly closed containers away from acids and acid fumes. (Refer to 4.3.2.)

5.5.7 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices. (Refer to 4.4.)

Calcium oxide meet the criteria in 40 CFR, Section 261.22(b) for a solid corrosive hazardous waste.

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5.6 Name. CUPRIC OXIDE, TECHNICAL CuO FW 79.54
Natural tenorite
Black copper oxide CAS 1317-38-0
Copper (II) oxide

5.6.1 Technical Description. Cupric oxide (pure) exists as black monoclinic crystals with a density of 6.3 - 6.49 g/cm^3. Its melting point is 1328°C. It is soluble in dilute acids, alkaline cyanides, and ammonium carbonate solutions. It is slowly soluble in ammonia, more readily if NH_4Cl is present. It is insoluble in hot or cold water.

5.6.2 Specification. Military, MIL-C-13600, Cupric Oxide.

5.6.3 Requirements. The Military Specification chemical and granulation requirements for cupric oxide, technical, are shown in Table VIII and Table IX.

TABLE VIII. Cupric oxide, technical - chemical requirements.

Requirement	% by wt
Assay (CuO), min	95.0
Moisture content, max	0.2
Acid insoluble matter, max	0.5

TABLE IX. Cupric oxide, technical - granulation requirements.

U S Standard Sieve 1/	% by wt
Thru 75 micrometer (No. 200) min	95

1/ Standard sieve designation in accordance with ASTM E 11.

5.6.4 Use. Cupric oxide, technical, is intended for use in pyrotechnic compositions.

5.6.5 Safety. Cupric oxide is irritating to the eyes and upper respiratory tract. Inhalation of dust shall be avoided. Contact with the eyes shall be avoided. Cupric oxide shall be used with adequate ventilation. The TLV for copper fume is 0.2 mg/m^3 and for dust and mists, as Cu, is 1 mg/m^3. The PEL for copper fume is 0.1 mg/m^3 and for dust and mist, it is 1 mg/m^3. (Refer to 4.3.1.)

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5.6.6 Storage. Cupric oxide shall be stored in a dry, cool place in tightly closed containers. (Refer to 4.3.2.)

5.6.7 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices. (Refer to 4.4.)

An EPA Hazardous Waste Classification is not listed in 40 CFR.

5.7 Name. LITHIUM HYDROXIDE, MONOHYDRATE, TECHNICAL
$LiOH.H_20$ FW 41.96

CAS 1310-66-3

5.7.1 Technical description. Lithium hydroxide, monohydrate (pure), exists as white monoclinic crystals with a density of 1.51 g/cm^3. Its solubility in water is 22.3 g/100 cm^3 at 10°C and 26.8 g/100 cm^3 at 80°C. It is slightly soluble in alcohol and insoluble in ether. Its heat of formation is - 188.9 K cal/mole at 25°C. Its heat of solution is -0.867 K cal/mole at 25°C and its specific heat is 0.453 cal/g/°C at 25°C. It loses water at 100 - 110°C.

5.7.2 Specification. Manufacturer's requirements. (No Government specification.)

5.7.3 Requirements. Lithium hydroxide, monohydrate, technical, free flowing grade, is commercially available with typical chemical analysis and specifications as shown in Table X and granulation requirements as shown in Table XI.

TABLE X. Lithium hydroxide, monohydrate, technical - typical chemical analysis and specification requirements.

Requirement	% by wt	
	Typical Analysis	Specification
Lithium hydroxide, monohydrate, free flowing grade (as LiOH)	57.2	56.5, min

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TABLE X. Lithium hydroxide, monohydrate, technical - typical chemical analysis and specification requirements (continued).

Requirement	% by wt	
	Typical Analysis	Specification (max)
Limits of Impurities		
Carbon dioxide (CO_2)	0.28	0.35
Chloride (Cl)	0.001	0.003
Sulfates (SO_4)	0.01	0.05
Calcium oxide (CaO)	0.01	0.03
Iron oxide (as FeO_3)	0.001	0.003
Sodium hydroxide (NaOH)	0.01	0.05
Insolubles	0.002	0.005

TABLE XI. Lithium hydroxide, monohydrate, technical - granulation requirements.

U S Standard Sieve 1/	% by wt
Thru 850 micrometer (No. 20)	100
Thru 180 micrometer (No. 80), min	20
Thru 150 micrometer (No. 100), min	10

1/ Standard sieve designation in accordance with ASTM E 11

5.7.4 Use. Lithium hydroxide, monohydrate, technical, is used in the production of multi-purpose lubricating grease, as an additive to the KOH electrolyte of alkaline type storage batteries, in preparation of anhydrous lithium hydroxide used as an absorbent for CO_2 in the air purification systems of submarines and space capsules, as an alkaline reagent for corrosion inhibition in steam boilers and a starting material in industrial chemical manufacturing. It is also used in water-based alkyd resin paints using lithium hydroxide dispersing agent, as a stabilizer in "single powder" photographic developers, photographic compositions, products and processes and as a component of copper electroplating baths.

5.7.5 Safety. Lithium hydroxide, monohydrate, is noncombustible and relatively nontoxic. It is caustic and therefore corrosive. The fact that lithium hydroxide, monohydrate, is a caustic material is the chief hazard. Its action on tissue can cause

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burns with possible ulceration and ultimate scarring. Prolonged contact with solutions of lithium hydroxide, monohydrate, has a destructive effect on tissue. Contact with the eyes, either in solid or solution form, can cause damage to the delicate tissue. Inhalation of the dust or mist can cause damage to the upper respiratory tract and to lung tissue, depending on the length of the exposure. Inhalation of dust or mist shall be avoided. Contact with the eyes, skin and clothing shall be avoided. Lithium hydroxide, monohydrate, shall be used with adequate ventilation. There is no TLV or PEL referenced for lithium hydroxide, monohydrate.

Avoid contact of lithium hydroxide, monohydrate, with concentrated acids. Concentrated acids react with the evolution of heat and spattering.

5.7.6 Storage. Lithium hydroxide, monohydrate, shall be stored in a cool, dry, well-ventilated place in tightly closed containers away from acids. (Refer to 4.3.2.)

5.7.7 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices. (Refer to 4.4.)

Lithium hydroxide, monohydrate, meets the criteria in 40 CFR, Section 261.22(b) for a solid corrosive hazardous waste.

5.8 Name. LITHIUM HYDROXIDE, TECHNICAL LiOH FW 23.95
Lithium hydrate

CAS 1310-66-3

5.8.1 Technical description. Lithium hydroxide (pure) exists as white tetragonal crystals with a density of 2.54 g/cm^3. Its melting point is 450°C and its boiling point is 924°C with decomposition. Its solubility in water is 12.8 g/100 cm^3 at 20°C and 17.5 g/100 cm^3 at 100°C. It is slightly soluble in alcohol and insoluble in ether.

5.8.2 Specification. Military, MIL-L-20213, Lithium Hydroxide (LiOH), Technical.

5.8.3 Requirements. The Military Specification chemical and granulation requirements for lithium hydroxide, technical, are shown in Table XII and Table XIII.

TABLE XII. Lithium hydroxide, technical - chemical requirements.

Requirements	% by wt
Assay (LiOH), min	97.6
Moisture Content, max	0.50

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TABLE XII. Lithium hydroxide, technical - chemical requirements (continued).

Requirements	% by wt
Max Limits of Impurities	
Sulfates and carbonates (as Li_2SO_4 or Li_2CO_3)	1.94 (total)
Calcium oxide (CaO)	0.20
Heavy metals	0.002
Chlorides	Trace
Sodium and potassium (Na and K)	0.20 (total)

TABLE XIII. Lithium hydroxide, technical - granulation requirements.

U S Standard Sieve 1/	Material Retained % by wt
4.75 millimeter (No. 4)	0 to 5
3.35 millimeter (No. 6)	25 to 70
2.36 millimeter (No. 8)	20 to 65
1.40 millimeter (No. 14)	5 to 45
1.18 millimeter (No. 16)	0 to 3
125 micrometer (No. 120)	0 to 1.5
Pan	0 to 0.5

1/ Standard sieve designation in accordance with ASTM E 11

5.8.4.1 Qualification. The lithium hydroxide furnished under this specification shall be, or qualified for listing, on the applicable qualified products list at the time set for opening of bids.

5.8.4.2 Materials. The lithium hydroxide shall be manufactured from lithium hydroxide monohydrate.

5.8.4.3 Solubility. Lithium hydroxide shall be completely soluble, giving a clear colorless solution without any sediment when tested as specified in MIL-L-20213E(SH).

5.8.4.4 Apparent density. The apparent density of the lithium hydroxide shall not be less than 0.38 g/cm^3 when tested as specified in MIL-L-20213E(SH).

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5.8.4.5 Hardness. Lithium hydroxide shall not be less than No. 95 hardness, and the amount retained on the No. 14 U S Standard Sieve screen at the end of the test shall not be less than 45 g when tested as specified.

5.8.4.6 Absorption. Initial breakthrough of detectable carbon dioxide in sample effluent must not occur for a minimum of 30 minutes when tested as specified. The absorption of carbon dioxide shall be not less that 0.70 g. of carbon dioxide per gram of lithium hydroxide at an absorption efficiency of 50 percent, determined as specified.

5.8.5 Use. Lithium hydroxide, technical, is intended for military use as a carbon dioxide absorbent on nuclear powered submarines. Its use is primarily as a back-up system for emergency absorbent requirements. It is also used in organic synthesis, as a catalyst in making alkyd resins and in esterification reactions and as a stabilizer in "single powder" photographic developers.

5.8.6 Safety. Lithium hydroxide is noncombustible and relatively non-toxic. It is caustic and therefore corrosive. The fact that lithium hydroxide is a caustic material is the chief hazard. Its action on tissue can cause burns with possible ulceration and ultimate scarring. Prolonged contact with solution has a destructive effect on tissue. Contact with the eyes, either in solid or solution form, can cause damage to the delicate tissue. Inhalation of the dust or mist can cause damage to the upper respiratory tract and to lung tissue, depending on the severity of the exposure. Inhalation of dust or mist shall be avoided. Contact with the eyes, skin, and clothing shall be avoided. Lithium hydroxide shall be used with adequate ventilation. There is no TLV or PEL referenced for lithium hydroxide. (Refer to 4.3.1.)

Avoid contact of lithium hydroxide with concentrated acids. Concentrated acids react with the evolution of heat and spattering.

5.8.7 Storage. Lithium hydroxide shall be stored in a cool, dry, well-ventilated place in tightly closed containers away from acids. (Refer to 4.3.2.)

5.8.8 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices. (Refer to 4.4.)

Lithium hydroxide meets the criteria in 40 CFR, Section 261.22(b) for a solid corrosive hazardous waste.

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5.9 Name. POTASSIUM HYDROXIDE, SOLUTION, TECHNICAL FW 56.11
Caustic potash, liquid

CAS 1310-58-3

5.9.1 Technical description. Potassium hydroxide, solution (30%, 34%, and 50% solution) is a colorless to pale yellow liquid. A 30% solution has a specific gravity of 1.2836 (H_2O=1). A 34% solution has a specific gravity of 1.3254 (H_2O=1) and a 50% solution has a specific gravity of 1.5050 (H_2O=1). The solubility of potassium hydroxide in water is 107 g/100 cm^3 at 15°C and 178 g/100 cm^3 at 100°C. The solute has a molecular weight of 56.11 g/mol.

5.9.2 Specification. Military, MIL-L-11751, Potassium Hydroxide Solution.

This specification covers three classes of potassium hydroxide solution as follows:

Class 1 - 34 percent solution
Class 2 - 50 percent solution
Class 3 - 30 percent solution

5.9.3 Requirements. The Military Specification requirements for the three classes of potassium hydroxide solutions are as shown in Table XIV, Table XV, and Table XVI.

TABLE XIV. Potassium hydroxide, solution (class 1 - 34% solution) - chemical requirements.

Requirements	% by wt	
	Min	Max
Potassium hydroxide (KOH)	32.0	36.0
Potassium carbonate (K_2CO_3)	---	3.0

TABLE XV. Potassium hydroxide, solution (class 2 - 50% solution) - chemical and physical requirements.

Requirements	% by wt	
	Min	Max
Potassium hydroxide (KOH)	45.0	50.0
Potassium carbonate (K_2CO_3)	--	3.0
Specific gravity at 15°/4°C	1.455	1.515

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TABLE XVI. Potassium hydroxide, solution (class 3 - 30% solution) - chemical requirements.

Requirements	% by wt	
	Min	Max
Potassium hydroxide (KOH)	29.5	30.5
Potassium carbonate (K_2CO_3)	---	0.90
Chloride (Cl)	---	0.005
Nitrogen compounds (as N)	---	0.001
Phosphate (PO_4)	---	0.0005
Sulfate (SO_4)	---	0.002
Ammonium hydroxide precipitate	---	0.01
Heavy Metals	---	0.002
Iron (Fe)	---	0.0005
Sodium (Na)	---	0.02

Potassium hydroxide solution, technical, is also available commercially in standard grade and mercury cell grade solutions of 45.0% and 50.0%. The standard method for chemical analysis of potassium hydroxide (caustic potash) 45% liquor is given in ASTM E 291.

5.9.4 Use. Potassium hydroxide solutions are intended for military use as desired by the procuring service. Commercially, it is used in producing soap, glass, textiles, dyes, organic and inorganic potassium compounds, electroplating and alkaline storage batteries.

5.9.5 Safety. Potassium hydroxide solutions are corrosive and a primary irritant. Concentrated solutions are strongly alkaline and are irritating to the eyes, skin and mucous membranes. Inhalation of mist can cause injury to the entire respiratory tract. In case of dilute solutions, symptoms of irritation may not be apparent until some time has passed. A chronic dermatitis may follow repeated skin contact. Inhalation of mist shall be avoided. Contact with the eyes, skin and clothing shall be avoided. Potassium hydroxide solutions shall be used with adequate ventilation. The TLV (ceiling limit) for potassium hydroxide solution is 2 mg/m^3 as KOH. (Refer to 4.3.1.)

5.9.6 Storage. Potassium hydroxide solution shall be stored in a cool, dry place in tightly closed containers away from acids and easily ignitable materials. (Refer to 4.3.2.)

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5.9.7 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices. (Refer to 4.4.)

Potassium hydroxide is on the EPA's list of hazardous substances, and it is a toxic, corrosive substance.

5.10 Name. POTASSIUM HYDROXIDE, TECHNICAL KOH FW 56.11
Caustic potash
Potash hydrate CAS 1310-58-3

5.10.1 Technical description. Potassium hydroxide (pure) exists as white, rhombic, deliquescent crystals with a density of 2.044 g/cm^3. It has a melting point of 360.4°C $\pm$ 0.7°C and a boiling point of 1320° - 1324°C. Its solubility in water is 107 g/100 cm^3 at 15°C and 178 g/100 cm^3 at 100°C. It is very soluble in alcohol and insoluble in ether and ammonia.

5.10.2 Specification. Federal, O-P-566, Potassium hydroxide, technical.

This specification covers four classes of potassium hydroxide, technical, as follows:

Class 1 - Flakes
Class 2 - Lumps
Class 3 - Pellets
Class 4 - Sticks

5.10.3 Requirements. The Federal Specification chemical and form requirements for potassium hydroxide, technical, are as shown in Table XVII and Table XVIII, respectively.

TABLE XVII. Potassium hydroxide, technical - chemical requirements.

Requirement	% by wt
Potassium hydroxide (KOH), min	85
Carbonate (as K_2CO_3), max	3

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TABLE XVIII. Potassium hydroxide, technical - form requirements.

Class	Form	Dimensions
1	Flakes	----------
2	Lumps	Larger than 12.5 millimeter (1/2-inch) and smaller than 50 millimeter (2 inches).
3	Pellets	6.3 millimeter (1/4-inch) maximum for any dimension
4	Sticks	12.5 millimeter (1/2-inch) maximum diameter

5.10.3.1 Manufacture. Potassium hydroxide shall be manufactured by the electrolytic method. The contractor shall certify that this requirement has been met.

5.10.3.2 Color, form and contamination. Potassium hydroxide shall be visibly free from foreign matter, shall have a white or pale yellow color, and shall have a form conforming to Table XVIII.

5.10.4 Use. Potassium hydroxide, technical, is intended for use in gas absorption and electroplating.

5.10.5 Safety. Potassium hydroxide is a corrosive solid and a primary irritant. It is nonflammable. Solid caustic and concentrated solutions are strongly alkaline and are irritating to the eyes, skin and mucous membranes. Inhalation of dust or mist can cause injury to the entire respiratory tract. In case of dilute solutions, symptoms of irritation may not be apparent until some time has passed. A chronic dermatitis may follow repeated skin contact. Inhalation of dust or mist shall be avoided. Potassium hydroxide shall be used with adequate ventilation. The TLV (ceiling limit) for potassium hydroxide is 2.0 mg/m^3. (Refer to 4.3.1.)

5.10.6 Storage. Potassium hydroxide shall be stored in a cool, dry place in tightly closed containers. (Refer to 4.3.2.)

5.10.7 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices. (Refer to 4.4.)

Potassium hydroxide is on EPA's list of hazardous substances. Potassium hydroxide is a toxic substance that meets the criteria in 40 CFR, Section 261.22(b)

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5.11 Name. SODIUM HYDROXIDE, SOLUTION, TECHNICAL FW 40.00
Caustic soda

CAS 1310-73-2

5.11.1 Technical description. Sodium hydroxide, solution (50%), is a colorless liquid. It has a boiling point of 142° - 148°C (288° - 298°F), a specific gravity of 1.525 at 20°C (68°F) (H_2O=1) and is miscible with water in all proportions. The solute has a molecular weight of 40.00 g/mol.

Sodium hydroxide, solution, technical, is available in a 50% commercial grade and in 50% and 73% rayon grades. The 50% commercial grade is a colorless, clear to hazy liquid with a density of approximately 1510 kg/m^3 at 38°C. The 50% rayon grade sodium hydroxide solution is a clear, colorless liquid with a density of approximately 1510 kg/m^3 at 38°C. The 73% rayon grade technical sodium hydroxide solution is a clear, colorless liquid with a density of approximately 1712 kg/m^3 at 80°C.

5.11.2 Specification. Manufacturer's requirements. (No Government specification.)

5.11.3 Requirements. The specification requirements of the 50% commercial grade are as shown in Table XIX. The specifications and typical analyses for 50% and 73% rayon grades are as shown in Table XX and Table XXI.

TABLE XIX. Sodium hydroxide (50% solution) technical, - specification requirements.

Requirements	% by wt	
	min	max
Sodium hydroxide (NaOH)	49.0	52.0
Maximum Limits of Impurities		
Sodium oxide (Na_2O)	38.0	40.4
Sodium carbonate (Na_2CO_3)		0.2
Sodium chloride (NaCl)		1.1
Iron (Fe)		0.0007
Sodium chlorate ($NaClO_3$)		0.4
Sodium sulfate (Na_2SO_4)		0.02
Silicon dioxide (SiO_2)		0.04
Aluminum oxide (Al_2O_3)		0.004

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TABLE XIX. Sodium hydroxide (50% solution) technical, - specification requirements (continued).

Requirements	% by wt	
	min	max
Calcium oxide (CaO)		0.006
Magnesium oxide (MgO)		0.002
Maganese (Mn)		0.0001
Nickel (Ni)		0.0002
Copper (Cu)		0.0002
Mercury (Hg)		0.00005

TABLE XX. Sodium hydroxide (50% solution - rayon grade), technical - typical analysis and specification requirements.

Requirements	% by wt		
	Typical Analysis	Specification	
		min	max
Sodium hydroxide (NaOH)	50.8	50.0	52.0
Maximum Limits of Impurities			
Sodium oxide (Na_2O)	39.4	38.7	40.3
Sodium carbonate (Na_2CO_3)	0.02		0.06
Sodium chloride (NaCl)	0.0006		0.003
Iron (Fe)	0.0008		0.0002
Sodium chlorate ($NaClO_3$)	0.00005		0.0001
Sodium sulfate (Na_2SO_4)	0.0005		0.002
Silicon dioxide (SiO_2)	0.0009		0.002
Aluminum oxide (Al_2O_3)	0.0001		0.0006
Calcium oxide (CaO)	0.001		0.002
Magnesium oxide (MgO)	0.000		0.0002
Manganese (Mn)	0.00002		0.00004
Nickel (Ni)	0.00001		0.00003
Copper (Cu)	0.00000		0.00005
Mercury (Hg)	0.00002		0.00005

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TABLE XXI. Sodium hydroxide (73% solution - rayon grade), technical - typical analysis and specification requirements (continued).

Requirements	% by wt		
	Typical Analysis	Specification	
		min	max
Sodium hydroxide (NaOH)	72.70	72.0	74.0
Maximum Limits of Impurities			
Sodium oxide (Na_2O)	56.35	55.83	57.35
Sodium carbonate (Na_2CO_3)	0.02		0.06
Sodium chloride (NaC1)	0.0010		0.004
Iron (Fe)	0.00013		0.00035
Sodium sulfate (Na_2SO_4)	0.0005		0.0020
Silicon dioxide (SiO_2)	0.001		0.0025
Aluminum oxide ($A1_2O_3$)	0.0002		0.0005
Calcium oxide (CaO)	0.0020		0.003
Magnesium oxide (MgO)	0.000		0.0003
Manganese (Mn)	0.00003		0.00004
Nickel (Ni)	0.00004		0.00005
Copper (Cu)	0.00000		0.00007

The standard method for chemical analysis of sodium hydroxide (caustic soda) is given in ASTM E 291.

5.11.4 Use. Sodium hydroxide, solution, technical, is an inexpensive alkali used in many chemical industries; it is used for effluent neutralization in pollution control, in the treatment of metals, and in production of alumina from bauxite. It is used as a reagent in paper production, the reclaiming of rubber and the manufacture of soap and synthetic detergent. It is also used as a reagent in the processing of rayon, mercerized cotton and synthetic fibers and in the decarbonizing of wool.

5.11.5 Safety. Sodium hydroxide solutions are corrosive and a primary irritant. Concentrated solutions are strongly alkaline and are irritating to the eyes, skin and mucous membranes. Inhalation of mist can cause injury to the entire respiratory tract. In case of dilute solutions, symptoms may not be apparent until some time as passed. A chronic dermatitis may follow repeated skin contact. Inhalation of mist shall be avoided. Contact with the eyes, skin and clothing shall be avoided. Sodium hydroxide

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solutions shall be used with adequate ventilation. The TLV (ceiling limit) for sodium hydroxide solution is 2 mg/m^3. (Refer to 4.3.1.)

Sodium hydroxide solution will react with water and acids to generate a considerable amount of heat; boiling and spattering of hot sodium hydroxide solution may result. The solution can react violently or explosively with any organic chemicals. Sodium hydroxide solution can react with some metals to form flammable hydrogen gas.

5.11.6 Storage. Sodium hydroxide solution shall be stored in a cool, dry, well-ventilated place in a tightly closed container away from acids. (Refer to 4.3.2.)

5.11.7 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices. (Refer to 4.4.)

Sodium hydroxide is on EPA's list of hazardous substances. Sodium hydroxide solutions are both toxic and corrosive.

5.12 Name. SODIUM HYDROXIDE, TECHNICAL NaOH FW 40.00
Caustic soda
Sodium hydrate CAS 1310-73-2
White caustic
Lye

5.12.1 Technical description. Sodium hydroxide (pure) exists as white, deliquescent crystals with an index of refraction of 1.3576 and a density of 2.130 g/cm^3. Its melting point is 318.4°C and boiling point is 1390°C. Its solubility in water is 42.0 g/100 cm^3 at 0°C and 347.0 g/100 cm^3 at 100°C. It is very soluble in alcohol and glycerine and insoluble in acetone and ether.

5.12.2 Specification. Federal, O-S-598, Sodium Hydroxide, Technical.

This specification covers four types of sodium hydroxide, technical grade, as follows:

Type I - Flake
Type II - Lump
Type III - Powder
Type IV - Bead

5.12.3 Requirements. The Federal Specification requirements for sodium hydroxide, technical, are as shown in Table XXII.

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TABLE XXII. Sodium hydroxide, technical - chemical requirements.

Requirements	% by wt
Assay (as NaOH), min	96.0
Total alkalinity (as Na_2O), min	74.4
Maximum Limits of Impurities	
Carbonate (as Na_2CO_3)	2.0

Form. Sodium hydroxide shall be flakes, lumps, powder or beads in accordance with the type of material specified.

Sodium hydroxide is also covered by ASTM D 456, Standard Specification for Caustic Soda (Anhydrous). The chemical requirements as specified are the same as those given in Table XXII.

Sodium hydroxide, technical, is commercially available in several grades and forms. It is available from one manufacturer in granular (regular and fine) and flake form. The specifications and typical granulation requirements are shown in Table XXIII and Table XXIV.

TABLE XXIII. Sodium hydroxide, technical - specification requirements.

Requirements	% by wt
Assay (as NaOH), min	97.0
Sodium oxide (Na_2O), min	75.1
Maximum Limits of Impurities	
Sodium carbonate (as Na_2CO_3), max	1.0
Sodium chloride (NaCl), max	1.0
Iron (Fe), max	0.003

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TABLE XXIV. Sodium hydroxide, technical - typical granulation requirements.

US Standard Sieve 1/	% by wt Retained		
	Granular	Fine Granular	Flake
50 millimeter (2 inch)	---	---	5.0
3.35 millimeter (No.6)	0.5	0.0	---
2.36 millimeter (No. 8)	---	---	85.0
2.00 millimeter (No. 10)	---	10.0	---
1.70 millimeter (No. 12)	70.0	---	95.0
850 micrometer (No. 20)	99.0	---	99.0
300 micrometer (No. 40)	99.5	99.0	---

1/ Standard sieve designation in accordance with ASTM E 11.

Sodium hydroxide, technical (rayon grade) is commercially available in granular (regular and fine) and flake form. The specification requirements are shown in Table XXV. The typical granulation requirements are the same as shown in Table XXIV.

TABLE XXV. Sodium hydroxide, (rayon grade), technical - specification requirements.

Requirements	% by wt
Assay (as NaOH), min	98.0
Sodium oxide (Na_2O), min	75.9
Limits of Impurities	
Sodium carbonate (as Na_2CO_3), max	0.5
Sodium sulfate (Na_2SO_4),max	0.1
Sodium chloride (NaCl), max	0.01
Sodium chlorate ($NaClO_3$)	<0.0001
Iron (Fe), max	0.002

Sodium hydroxide (reactor grade) is commercially available in course granules and thick plate - type flake form The specification requirements for the course granules and thick plate - type flakes are the same as those given in Table XXV.

The typical granulation requirements are as shown in Table XXVI.

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TABLE XXVI. Sodium hydroxide, (reactor grade), technical - typical granulation requirements.

U S Standard Sieve 1/	% by wt Retained	
	Flake	Granular
50 millimeter (2 inch)	5.0	---
3.35 millimeter (No.6)	---	0.5
2.36 millimeter (No. 8)	85.0	---
1.70 millimeter (No. 12)	95.0	70.0
850 micrometer (No. 20)	99.0	99.0
300 micrometer (No. 40)	---	99.5

1/ Standard sieve designation in accordance with ASTM E 11

Sodium hydroxide, technical, (standard grade) is commercially available in flake, granular and beaded form. The specification requirements are shown in Table XXVII. The typical granulation requirements are as shown in Table XXVIII.

TABLE XXVII. Sodium hydroxide, (standard grade), technical - specification requirements.

Requirement	% by wt	
	Flake and Granular	Beads
Sodium hydroxide (equiv) (NaOH), min	96.0	96.0
Sodium oxide (equiv)(Na_2O), min	75.0	74.4
Maximum Limits of Impurities		
Sodium carbonate (as (Na_2CO_3)	3.0	1.60
Sodium chloride (NaCl)	1.7	2.20
Sodium sulfate (Na_2SO_4)	0.10	0.035 1/
Sodium chlorate ($NaClO_3$)	None	0.02
Silicon (Si)	0.02	0.003 1/

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TABLE XXVII. Sodium hydroxide, (standard grade), technical - specification requirements (continued).

Requirement	% by wt	
	Flake and Granular	Beads
Maximum Limits of Impurities		
Iron (Fe)	0.0025	0.002
Calcium (Ca)	0.0015	0.0010 1/
Magnesium (Mg)	0.002	0.0010 1/
Aluminum (Al)	0.001	0.0006 1/
Manganese(Mn)	0.0005	0.0004 1/
Copper (Cu)	0.0001	0.0002
Nickel (Ni)	0.0001	0.0004

1/ No specification. Represents typical data for beads.

TABLE XXVIII. Sodium hydroxide (standard grade), technical - typical granulation requirements.

U S Standard Sieve 1/	% by wt Retained		
	Flake	Granular	Beads
9.5 millimeter (3/8 inch)	1.0	---	---
6.3 millimeter (1/4 inch)	5.0	---	---
4.75 millimeter (No. 4)	15.0	0.0	---
2.36 millimeter (No. 8)	75.0	12.0	---
2.00 millimeter (No. 10)	---	27.0	0.0
1.40 millimeter (No. 14)	4.0	38.0	---
1.18 millimeter (No. 16)	0.0	10.0	---
850 micrometer (No. 20)	---	9.0	10.0-25.0
600 micrometer (No. 30)	---	4.0	---
300 micrometer (No. 40)	---	---	50.0-70.0
250 micrometer (No. 60)	---	---	10.0-15.0
Thru 250 micrometer (No. 60)	---	---	1.0-10.0
Pan	---	Trace	---

1/ Standard sieve designation in accordance with ASTM E 11.

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5.12.4 Use. Sodium hydroxide is intended for use in various cleaning, washing, and scouring processes, with or without soap, as conditions demand, and where a strongly alkaline material is desired.

5.12.5 Safety. Sodium hydroxide, both solid and in solution, has a markedly corrosive action upon all body tissue. Its corrosive action on tissue causes burns and frequently causes deep ulceration. Mists, vapors and dust of sodium hydroxide cause burns, and contact of either the solid or solution form with the eyes rapidly causes severe damage to the delicate tissue. Inhalation of the dust or concentrated mists can cause damage to the upper respiratory tract and to lung tissue; therefore, inhalation of dust or mists shall be avoided. Contact with the eyes, skin and clothing shall be avoided. Sodium hydroxide shall be used with adequate ventilation. The TLV (ceiling limit) for sodium hydroxide is 2 mg/m^3. (Refer to 4.3.1.)

Sodium hydroxide will react with water and acids to generate a considerable amount of heat; boiling and spattering of hot sodium hydroxide solution may result.

5.12.6 Storage. Sodium hydroxide shall be stored in a cool, dry, well-ventilated place in tightly closed containers away from acids. (Refer to 4.3.2.)

5.12.7 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices. (Refer to 4.4.)

Sodium hydroxide is on EPA's list of hazardous substances. Sodium hydroxide is a toxic, corrosive substance.

5.13 Name. STANNIC OXIDE, TECHNICAL SnO_2 FW 150.69
Tin (IV) oxide
Stannic anhydride CAS 18282-10-5
Tin dioxide

5.13.1 Technical description. Stannic oxide (pure) exists as white tetragonal (also hexagonal or rhombic) crystals with a density of 6.95 g/cm^3. Its melting point is 1127°C and it sublimes at 1800°-1900°C. It is soluble in concentrated sulfuric acid and hydrochloric acid. It is insoluble in hot and cold water.

5.13.2 Specification. Military, MIL-S-50005, stannic oxide, technical.

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This specification covers two classes of stannic oxide, technical:

Class 1
Class 2

5.13.3 Requirements. The Military Specification chemical and granulation requirements for stannic oxide, technical, are shown in Tables XXIX and XXX.

TABLE XXIX. Stannic oxide, technical - chemical requirements.

Assay (as SnO_2 on a dry basis), min	98.2
Maximum Limits of Impurities	
Volatile content	0.50
Acidity (as H_2SO_4)	0.01

TABLE XXX. Stannic oxide, technical - granulation requirements.

U S Standard Sieve 1/	% by wt	
	Class 1	Class 2
Thru 75 micrometer (No. 200) min	95	95
Thru 45 micrometer (No. 325) min	85	90

1/ Standard sieve designation in accordance with ASTM E 11.

5.13.4 Use. Stannic oxide, technical, is intended for use as an antifouling agent.

5.13.5 Safety. Stannic oxide has a low toxicity and is nonflammable. Inhalation of dust or fumes of inorganic tin is known to cause a benign pneumoconiosis. It produces distinctive progressive X-ray changes of the lungs as long as exposure persists, but there is no distinctive fibrosis, no evidence of disability and no special complicating factors. Inorganic tin compounds may cause mild irritation to the skin and mucous membranes. Inhalation of dust shall be avoided. Contact with the eyes, skin and clothing shall be avoided. Stannic oxide shall be used with adequate ventilation. The TLV for tin oxide and inorganic compounds as Sn is 2 mg/m^3. There is no PEL. (Refer to 4.3.1.)

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5.13.6 Storage. Stannic oxide shall be stored in a cool, dry, well-ventilated place in tightly closed containers. (Refer to 4.3.2.)

5.13.7 Disposal. For appropriate procedures, contact the Installation Environmental Office, the DRMO, or Safety and Health Offices. (Refer to 4.4.)

An EPA Hazardous Waste Classification is not listed in 40 CFR.

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and add 10 ml of 1N sodium hydroxide solution. When solution is com-

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Subject term (key word) listing.

Exposure limits, hazardous chemicals
Ammonium hydroxide
Barium hydroxide
Barium oxide
Calcium hydroxide
Calcium oxide
Lithium hydroxide
Lithium hydroxide, monohydrate
Potassium hydroxide
Potassium hydroxide, solution
Sodium hydroxide
Sodium hydroxide, solution
Stannic oxide
Safety, hazardous chemicals

6.2 Changes from previous issue. Asterisks or vertical lines are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Preparing activity: Army - EA

Project Number 6810-1288

Custodians:

Army - EA
Navy - SH
Air Force - 68

Review activities:

Army - AR, MD, SM
Navy - MS, OS, YD

User activities:

Army - ME
DOD -- DP

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